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CSA0702-Computer Networks

Role of the OSI Model in Software-Defined Networking (SDN)

Guide By:

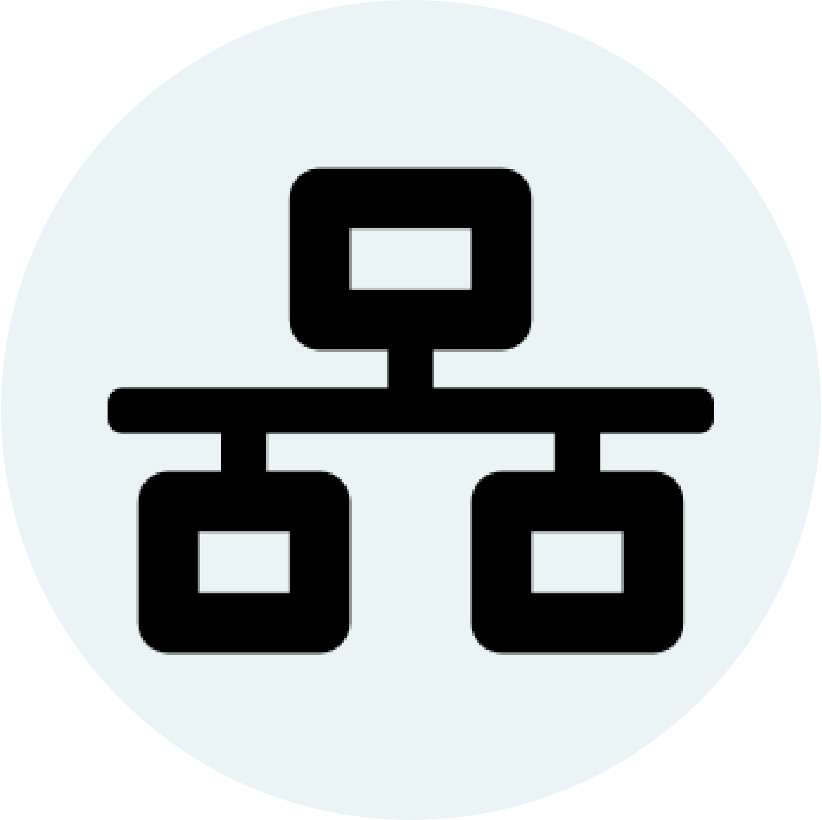
Dr.R.Senthilselvan

Presented By:

Sanjai V (192312486)

Introduction

- Modern networks are becoming larger, more dynamic, and harder to manage using traditional methods.
- Software-Defined Networking (SDN) separates the network's control logic from the underlying hardware.
- The OSI Model provides the layered reference structure that SDN relies on to organize network functions.
- Understanding how OSI layers map to SDN helps explain how modern networks are designed and controlled.



What is the OSI Model?

The Open Systems Interconnection (OSI) Model is a conceptual framework that divides network communication into **seven layers**, each responsible for a specific function.

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

Key Idea

Each layer communicates only with the layers directly above and below it, allowing systems to be built and updated independently.

What is Software-Defined Networking (SDN)?

Software-Defined Networking (SDN) is an approach to networking that separates the **control plane** (decision-making) from the **data plane** (packet forwarding), allowing centralized, programmable control of the network.

Key Features

- Centralized network control
- Programmable network behavior
- Dynamic traffic management
- Separation of control and data planes

Role of the OSI Model in SDN

SDN does not replace the OSI Model — it reorganizes how OSI layer functions are controlled. The OSI Model still explains *what* each layer does, while SDN changes *how* those functions are managed.

- Data Link & Network layers: SDN switches handle forwarding based on rules set by a centralized controller.
- Network layer (Layer 3): SDN controllers make routing decisions instead of each device deciding independently.
- Transport & Session layers: SDN can monitor flows and apply traffic engineering across these layers.
- Application layer: Northbound APIs let applications request network behavior directly from the controller.

Advantages

Combining the OSI Model's structure with SDN's centralized control brings several benefits to modern networks.

- Simplified network management
- Improved scalability
- Better resource utilization
- Reduced operational cost
- Faster deployment of network policies
- Centralized traffic monitoring
- Flexible, programmable networks
- Consistent policy enforcement

Real-World Applications

SDN and OSI-based network design are used across many modern networking environments.

- Google and large-scale data centers
- Cloud computing infrastructure
- 5G and mobile network cores
- Enterprise campus networks
- Internet Service Providers (ISPs)
- Network security and firewalls

Challenges

Despite its benefits, applying SDN across OSI layers introduces some practical challenges.

- Single point of failure at the controller
- Security risks from centralized control
- Complex migration from traditional networks
- Interoperability across different vendors
- Skilled professionals required for management

Future Scope

The role of the OSI Model in SDN will continue to expand as networks grow more automated and intelligent.

- AI-driven SDN controllers
- Integration with 6G networks
- Self-healing and self-optimizing networks
- Intent-based networking
- Greater automation across all OSI layers

Conclusion

- The OSI Model provides the structural foundation that explains how network communication works layer by layer.
- SDN reshapes how those layers are controlled by centralizing decision-making and enabling programmability.
- Together, the OSI Model and SDN form the basis of modern, flexible, and efficient network architectures.

References

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THANK YOU!